

CONTACTS

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PROFILE

I am a **Lecturer in Chemical Engineering** at the **University of Edinburgh** and the Principal Investigator of the M2D2 group (Molecular Modelling and Data-Driven Materials Design). My expertise lies in developing **multiscale and multi-method modelling strategies**, integrating molecular, thermodynamics, and data-driven techniques, to tackle challenges in designing **innovative materials and processes for environmental applications**, such as carbon capture, hydrogen management, biomaterials, membrane separations.

RESEARCH EXPERIENCE

12/2023 – Present **Assistant Professor (Lecturer) in Chemical Engineering**

Institute for Materials & Processes, School of Engineering.
University of Edinburgh, UK.



11/2021 – 11/2023 **Marie Curie Postdoctoral Research Fellow**

National Centre for Scientific Research Demokritos, Athens, Greece
Inst. of Informatics and Telecommunications. Project ML-MULTIMEM:

“Machine Learning-aided Multiscale Modelling Framework for Polymer Membranes”



11/2020 – 10/2021 **Postdoctoral Fellow, Chemical Engineering**

University of Bologna, Italy. Project MuMPol: *“Modelling and Design of Multiphase Polymeric Materials for High Performance Applications Across Multiple Scales”*. Funded by the Dutch Polymer Institute (DPI).



11/2019 – 10/2020 **Postdoctoral Fellow, Chemical Engineering**

University of Bologna, Italy. Project: *“Modelling and characterization of sorption and transport of gases in polymeric membranes for gas separation”*.



INTERNATIONAL EXPERIENCE

09/2024 – 11/2024 **Visiting Researcher**. SINTEF Digital, Oslo, Norway. In the framework of the European Lighthouse to Manifest **Trustworthy and Green AI** Mobility Grant. Project *“Hybrid-AI Model for mixed-gas Sorption Upper Bounds”*.



SINTEF

11/2019 – 11/2019 **Visiting Researcher**.

National Centre for Scientific Research Demokritos, Athens, Greece. HPC Europa 3 Mobility Grant Recipient.



11/2019 – 11/2019 **Visiting PhD Student**.

National Technical University of Athens, Athens, Greece.



EDUCATION

- **Ph.D. in Chemical Engineering** 11/2016 – 10/2019
Department of Civil, Chemical, Environmental and Materials Engineering, University of Bologna, Italy
Thesis: “*Molecular and Thermodynamic Simulations for Pure and Mixed Gas Sorption in Polymeric Membranes*”. Degree obtained: March 2020.
- **Master’s Degree in Chemical and Process Engineering** 09/2014 – 10/2016
University of Bologna – Italy. 110/110, 1st class Honours.
Thesis: “*Study of membrane materials for CO₂/CH₄ separation with a combined macroscopic/molecular approach*”.
- **Bachelor’s Degree in Chemical and Biochemical Engineering** 09/2011 – 07/2014
University of Bologna – Italy. 110/110, 1st class Honours.
Thesis: “*Experimental characterization of ceramic materials for membrane contactors*” Work carried out in the framework of an **industrial partnership with Saipem S.p.A.**
- **High School Diploma, Foreign Languages Curriculum** 2006 – 2011
Liceo Classico “L.A. Muratori”, Modena – Italy. 100/100 and Honours.

TEACHING EXPERIENCE

- **Course Organiser**, “Batchwise and Semibatch processing”, Univ. of Edinburgh 2024/2025, 2025/2026
- **Adjunct Professor**, “Molecular Design and Materials Simulation” Univ. of Bologna, Italy 2020 – 2021
- **Teaching Assistant**, University of Bologna, Italy. 2016 – 2020
Thermodynamics for Chemical Engineering (BSc level), Thermodynamics for Chemical Engineering (MSc level), Introduction to Chemical Engineering, Fluid Dynamics
- **Summer School Instructor** 09/2019, 01/2020
“Blended Summer School on Engineering Processes, Systems & Materials”.
University of Bologna, Italy & University of Melbourne, Australia.

RESEARCH PROJECTS

- | | | |
|---------------|---|---|
| ■ 2026 | An Interdisciplinary Alliance on Materials for Clean Energy (ALMATCLEAN)
Una Europa Seed Funding. 38,683 €, Coordinating PI |  |
| ■ 2026 | Advancing materials for the energy transition with molecular precision
synthesis and in silico design. Royal Society of Edinburgh; £7,848, PI |  |
| ■ 2026 | Machine Learning-aided Optimisation of Composites Recycling
EPSRC Impact Acceleration Account; £7,077, PI |  |
| ■ 2025 | Reducing greenhouse gas emissions in clinical settings: investigating
nitrous oxide separation with membranes
Edinburgh Research Partnership in Engineering (ERPE); £4,000, PI |  |
| ■ 2024 - 2028 | CharMPol (<i>Characterization and Modeling of high-performance
polymers for gas barrier applications</i>)
Dutch Polymer Institute; 155,000 €, co-Principal Investigator. |  |
| ■ 09-11/2024 | ENFIELD: European Lighthouse to Manifest
Trustworthy and Green AI- Exchange scheme
European Commission, 7,200 € , Visiting researcher |   |

■ 2021-2023	ML-MULTIMEM (<i>Machine Learning-aided Multiscale Modelling Framework for Polymer Membranes</i>) Marie Skłodowska-Curie Actions Individual Fellowships European Commission, 153,085 € , Postdoctoral Fellow	
■ 2023	<i>A Machine Learning Platform for the design of Hydrogen-Ready materials</i> Royal Society of Edinburgh, Research Workshop Autumn 2022 £ 9,600 , co-Principal Investigator	
■ 2020 - 2024	MuMPol (<i>Modelling and Design of Multiphase Polymeric Materials for High Performance Applications Across Multiple Scales</i>) Dutch Polymer Institute; 140,000 € , co-Investigator.	
■ 2021	GHAIA (<i>Geometric and Harmonic Analysis with Interdisciplinary Applications</i>). European Commission, 12,000 € 6-month secondment at the Department of Chemical Engineering at the Massachusetts Institute of Technology (MIT). Due to travel restrictions resulting from the COVID-19 pandemic, it was transformed into a remote collaboration.	
■ 2019 - 2022	ISCRA (Italian Super Computing Resource Allocation) – CINECA Over 10 million CPU hours (multiple projects: GD-MOF, GS-SCP, MM-PMC, SimPIM, PermCTA)	
■ 2018 – 2019	HPC-EUROPA3- mobility programme towards High Performance Computing Centres in Europe. Award: 3,500 € + 300,000 CPU hours	

PUBLICATIONS LIST

Full updated list on Google Scholar: <https://scholar.google.it/citations?user=ax3B6GIAAAAJ&hl=en&oi=ao>

AWARDS AND SCHOLARSHIPS

- 2021 **PhD Thesis Award** “Excellence Award in Thermodynamics and Transport Properties 2021” from the European Federation of Chemical Engineering: 2nd place
“Patrick Meares” **PhD Thesis Award** from European Membrane Society: runner-up candidate (800€, as a travel grant to present my PhD Thesis at the Euromembrane 2021 conference)
- 2020 **Best presentation award** for the work “*Multicomponent effects in CO₂/CH₄ separation with glassy membranes*” at the Network Young MemBrains meeting, London, virtual 2020. (300€)
Travel Award from the European Membrane Society to attend ICOM 2020. (300€)
- 2019 **Travel Grant** from the Membrane Society of Australasia to attend IMSTEC 2020. (700€)
Travel Award from the European Membrane Society to attend EWM 2019. (650€)
- 2018 **Travel Award** from the European Membrane Society to attend Euromembrane 2018. (420€)
- 2016 Three-year **Scholarship** from the Italian Ministry of Education to pursue the Ph.D. in Chemical and Process Engineering at the University of Bologna (68,000€)
- 2014 **Scholarship**: Merit-based one-year full tuition fee waiver for the Master’s Degree in Chemical and Process Engineering at the University of Bologna, Italy. (1,500€)
- 2011 **Scholarship**: Merit-based one-year full tuition fee waiver for the Bachelor’s Degree in Chemical and Biochemical Engineering at the University of Bologna, Italy. (1,300€)

INVITED LECTURES AND PRESENTATIONS

- **Invited presentation at the CECAM workshop** Molecular simulations meet data science for polymer science. Manchester, 22-24/06/2026
- **Invited Seminar at the Una Europa Future Materials and Technologies Webinar Series.** Title: “*Combined physics-based and data-driven modelling strategies for the design of sustainable gas separations*”. 19/03/2026

- **Invited Seminar at the Warwick Centre for Predictive Modelling** Seminar Series. Title: “Multiscale and multi-method modelling strategies for polymer membranes”., Warwick, 16/02/2026
- **Invited Presentation at the New Starters in AI for Chemistry Symposium.** Title: “A Transfer Learning Approach to Predict Equation of State Parameters for Polymers”. Leeds, 27/01/2026
- **Invited Seminar at the Complex Network Analysis Group.** NCSR Demokritos, Greece. Title: “Convolutional Graph Neural Networks for Learning Interactions in Molecular Simulations”. [Link](#). 21/01/2023
- **Invited Seminar at the European Federation of Chemical Engineering Spotlight Talks Webinar Series.** Title: “Machine-Learning Methods to Facilitate Coarse-Grained Molecular Simulations”. [Link](#). +40 participants and ~200 views. 24/11/2022
- **Invited Seminar for the NanoAI Series,** NCSR Demokritos Athens, Greece. [Link](#). Title: “Coarse Grained Molecular Simulations Facilitated by Machine Learning. 20 participants. 21/10/2022.
- **Instructor, 57th NCSR Demokritos Summer School,** Athens, Greece. [Link](#). Title: “Empowering multi-scale simulations with Machine Learning”. +100 participants. 12/07/2022
- **Invited Seminar, Research and Industrial Seminars Series – University of Bologna,** Italy. Title: “Simulation Strategies for the Prediction of the Gas Separation Properties of Polymeric Membranes”. ~50 participants. [Link](#). 26/03/2021
- **Invited Lecture – University of Melbourne, Australia,** for the class “Recent Advances in Separation Processes”, Department of Chemical Engineering. Title: “Calculation of Solubility in Polymers with Equations of State and Non-Equilibrium models”. 30/01/2020

CONFERENCES AND EVENTS ORGANISATION

- **Conference Chair** of the CCP5 Annual General Meeting 2026, Edinburgh, UK, 9-11/09/2026
- **Organising Committee** – Non-Equilibrium Molecular Dynamics Conference (NEMD) and Summer School 2026, Edinburgh, UK, 6-10/07/2026
- **Scientific Committee** - European Symposium on Applied Thermodynamics, Edinburgh, UK, 9-12/06/2024
- **Organizing Committee and Event Co-chair** Artificial Intelligence in Natural Sciences and Technology Workshop, 12th Hellenic Conference on Artificial Intelligence, Corfu, Greece. 7-9/09/2022
- **Organising Committee,** Blended Summer School on Engineering Processes, Systems and Materials (EPSM), Bologna, Italy. 10-12/09/2019
- **Organising Committee:** International Workshop on Physics of Membrane Processes, Bologna, Italy 02/09/18

REVIEWING ACTIVITIES

- **Peer Reviewer** for several high profile Scientific Journals (Scientific Reports (Nature), Journal of Membrane Science, Fluid Phase Equilibria, Journal of Natural Gas Science and Engineering, Computational Materials Science (Elsevier), Journal of Chemical Physics (AIP Publishing), Physical Chemistry Chemical Physics (Royal Society of Chemistry)).
- **Expert Rapporteur** for Competitive Grant Schemes (EU, UKRI, ESF).

INDUSTRY ENGAGEMENT AND TRANSLATIONAL RESEARCH

- I am currently collaborating with the polymer recycling start-up “**Sustainable Extricko**” on optimizing process conditions via data-driven modelling, supported by EPSRC Impact Accelerator Account funding.
- I collaborated with **ExxonMobil** (Dr Bennett Marshall) on the extension of a modelling method for gas solubility in glassy polymers, which resulted in 1 publication and 1 manuscript in preparation.
- I have a longstanding involvement in projects funded by the industrial consortium **Dutch Polymer Institute** (MuMPol 2020-2024, CharmPol 2024-2028), with the aim of predicting gas transport properties of semicrystalline polymers via a combination of molecular modelling and macroscopic thermodynamic approaches. The research team periodically reports to and sets targets together with the participating

industrial partners: **Envalor, Shell, Saudi Aramco, Sabic, King Fa**. These projects have yielded 4 publications and the research has been disseminated yearly to the industrial and academic partners portfolio of DPI.

- As part of the ML-MULTIMEM postdoctoral project, I collaborated with **SciFY - Science for You**, a not-for-profit software company, on packaging and distributing the project's code adhering to high-quality standards for reusability and community impact.

COMMUNITY AND OUTREACH ACTIVITIES

- 2026 – present: I am participating as a **Mentor** in the European Federation of Chemical Engineers Early Careers Mentoring Programme.
- 2025 – present: **Vice-Chair** of the **Molly Ferguson Initiative**, which aims at championing women in engineering and promoting a supportive and inclusive environment within the School of Engineering at UoE.
- 2023 – 2024: Member of the **Researcher Mental Health Observatory (ReMO) Survey Special Interest Group**. I was Country Coordinator (Greece) for the dissemination of a now published (I am co-author of the work) large-scale benchmark survey of researchers' mental health in Europe.
- 2022 – 2023 : I am part of the **1000 pioneers for AI in Greece Network**, that organizes technical training and public outreach activities concerning the fundamentals of AI models, their strength and limitations, as well as their utility and relevance for different stakeholders (natural science and humanities researchers, students, NGOs & civil society...).
- 2018 – 2023: **Board Member** at **FAIL! – Inspiring Resilience**: Grass-root initiative aimed at increasing awareness about mental well-being in academia, by inviting professors to share experiences of challenging times from their professional or personal lives in a TED talk-style format. I held various executive roles within the association, and served as event organizer and chief of communication.
- 30/09/2022 : I participated in the **European Researchers' Night** at the National Technical University of Athens, Greece. Through computer games and interactive demonstrations, I interacted with children, high school students, and university students, to introduce them to the ideas of molecules, molecular simulations, and supervised machine learning.
- A.Y. 2020-2021 : I participated as a **Mentor in the MyPhDMentor program**, organized by ADI (Italian PhD students and PhD graduates' association).

NATIONALITY

- Italian

LANGUAGE SKILLS

- Italian: Native
- English: Proficient (CEFR C2)
- French: Intermediate (CEFR B2)
- German: Intermediate (CEFR B1)
- Greek: Basic (CEFR A2)

INFORMATICS SKILLS

- **Programming languages**: Python, Matlab, Fortran, C++
- **Machine Learning**: PyTorch, scikit-learn
- **Materials Simulation** software: LAMMPS, MAPS, Materials Studio, VMD, ASE
- Familiarity with Windows and **Linux** operating systems.
- Familiarity in the use of **HPC** environments, parallel code execution, **GPU** acceleration.
- Version control and collaborative code development (**Git**)
- Basic knowledge of **Engineering and Process Simulation** software: AutoCad, Phast, Aloha, Aspen Hysys, Ansys Fluent